

Beyond Conventional Sentiment Indicators: Bitcoin's Hidden Potential in VIX Forecasting

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ABSTRACT

This study introduces a novel decomposition framework aligned with U.S. stock market trading hours, identifying Bitcoin's overnight returns as a powerful indicator of investor sentiment. By incorporating Bitcoin's overnight returns into standard forecasting models, we observe a significant improvement in the predictability of equity market volatility dynamics across both in-sample and out-of-sample analyses. Furthermore, long-short trading strategies based on these enhanced forecasts consistently outperform conventional approaches, generating substantial economic gains. These findings suggest that existing volatility models could benefit from the inclusion of cryptocurrency market signals. Overall, our results highlight the growing integration of cryptocurrency and traditional financial markets.

Keywords: VIX; Bitcoin; Overnight Returns; Investor Sentiment

1. Introduction

The Chicago Board Options Exchange Volatility Index (VIX), commonly known as the “investor fear gauge,” serves as a crucial indicator of market sentiment. While existing literature examines historical patterns and equity market indicators for implied volatility (e.g., Harvey and Whaley, 1992; Christensen and Prabhala, 1998; Jiang and Tian, 2005; Bekaert and Hoerova, 2014; Bardgett et al., 2019), forecasting the VIX with accuracy remains a challenge for researchers, practitioners, and policymakers. Given Bitcoin’s 24/7 trading, it offers a valuable real-time indicator of market sentiment and volatility, even when equity markets are closed. Building on this insight, this study proposes a novel predictor for the VIX: Bitcoin returns during U.S. equity market non-trading hours, referred to as “Bitcoin overnight returns”.

Our study is motivated by fundamental theoretical insights. Bitcoin, characterized by “hard-to-value” fundamentals¹, is uniquely influenced by its mining-based ecosystem (Easley et al., 2019) and exhibits high sensitivity to shifts in investor sentiment (Conlon et al., 2024). Prior research (Berkman et al., 2012; Aboody et al., 2018) suggests that overnight stock returns reflect firm-specific investor sentiment. Extending this perspective to Bitcoin, we argue that its continuous trading shares characteristics with high-sentiment beta stocks, which are highly responsive to shifts in sentiment (Jo et al., 2020). Thus, we hypothesize that Bitcoin overnight returns capture market sentiment not yet reflected in equity market pricing.

To empirically test this hypothesis, we utilize five-minute Bitcoin price data from 2018 to 2023 and uncover three main findings. First, Bitcoin overnight returns exhibit patterns consistent with established sentiment measures, suggesting their potential as a new indicator of investor sentiment. Second, incorporating Bitcoin overnight returns significantly improves

¹ Related studies include Makarov and Schoar (2020), Chod and Lyandres (2021), Cong et al. (2021), Detzel et al. (2021), Gryglewicz et al. (2021), Biais et al. (2023), and Sockin and Xiong (2023).

both in-sample and out-of-sample predictive accuracy in VIX forecasting models. Third, trading strategies incorporating Bitcoin overnight returns generate economically significant gains compared to conventional approaches.

We first examine the relationship between Bitcoin overnight returns and investor sentiment. Using established sentiment identification methodologies (Barber et al., 2009; Berkman et al., 2012; Aboody et al., 2018), we sort Bitcoin overnight and trading-hour returns into decile portfolios based on abnormal investor attention.² Our analysis reveals overnight-to-trading-hour return reversals, with the magnitude of these reversals increasing substantially as investor attention rises. This variation in return reversals across attention-sorted portfolios provides compelling evidence that Bitcoin overnight returns potentially reflect retail investor sentiment.

Next, we investigate whether these sentiment-driven Bitcoin overnight returns hold predictive power for VIX dynamics. Our in-sample Vector Autoregressive (VAR) analysis shows that heightened sentiment, as reflected in Bitcoin overnight returns, predicts declines in the VIX during the subsequent trading session. In contrast, Bitcoin trading-hour returns lack significant predictive power for the VIX. This pattern suggests that Bitcoin overnight returns contain unique information relevant to VIX forecasting. Extending these in-sample results, we evaluate the out-of-sample predictive performance of Bitcoin overnight returns. Models incorporating Bitcoin overnight returns consistently outperform baseline models in forecasting VIX changes, even after controlling for established predictors, such as holiday effects and FOMC announcements. The superior forecasting performance is validated using Diebold-Mariano tests (Diebold and Mariano, 1995) and Model Confidence Set analysis (Hansen et al., 2011), confirming the significant predictive value of Bitcoin overnight returns

² Abnormal investor attention is defined as the level of investor interest that deviates significantly from the average. Specifically, we use two distinct attention proxies: Bitcoin’s unique active addresses and Google search volume, each normalized by their trailing monthly averages.

in enhancing VIX forecasting models.

To assess the economic significance of our findings, we implement long-short trading strategies using VIX derivatives. Our analysis shows that strategies exploiting Bitcoin overnight returns generate economically significant risk-adjusted returns relative to a passive buy-and-hold benchmark. The Heterogeneous Autoregressive (HAR) model, augmented with S&P 500 futures and Bitcoin overnight returns (HAR-ON), shows superior performance. We also find substantial economic benefits, as indicated by certainty equivalent returns, that persist across different levels of investor risk aversion. These findings highlight the unique insights provided by Bitcoin overnight returns, emphasizing their role in capturing crucial market information outside regular trading hours.

Furthermore, we examine the robustness of Bitcoin overnight returns' predictive power across multiple dimensions. First, we perform a temporal decomposition, comparing pandemic and non-pandemic subperiods to assess prediction consistency across different market regimes. Second, we evaluate predictive performance for alternative U.S. equity market implied volatility indices beyond the VIX, broadening the scope of our analysis. Third, we extend our framework to international markets, investigating the cross-border applicability of Bitcoin overnight returns in forecasting implied volatility. Across all these dimensions, our findings remain robust, reinforcing the validity of our results and demonstrating the widespread applicability of Bitcoin overnight returns as a significant predictor of implied volatility.

Our study contributes to several strands of literature. First, it offers novel insights into the relationship between cryptocurrency markets and conventional financial sentiment. By leveraging Bitcoin's unique characteristics—continuous trading and high retail investor participation (Kogan et al., 2024)—we identify a new channel through which sentiment affects market uncertainty. While conventional sentiment indicators typically operate at low

frequencies (Gao and Süß, 2015), recent studies have shown that overnight stock returns reflect firm-specific sentiment (Berkman et al., 2012; Aboody et al., 2018). We extend this framework by demonstrating that cryptocurrency overnight returns function as a market-wide sentiment indicator at a daily frequency. Our measure provides real-time, market-based information on investor sentiment, capturing additional insights outside regular trading hours and further enriching the literature on investor sentiment.

Second, our study enriches the literature on forecasting the VIX. While prior research has explored cryptocurrency's portfolio diversification benefits (Jiang and Zhang, 2023) and its sensitivity to investor sentiment (Kapar and Olmo, 2020; Lee and Jeong, 2023), the predictive power of cryptocurrencies for equity market volatility remains largely unexplored. Moreover, despite studies suggesting that trading strategies based on VIX forecasts are generally unprofitable (Konstantinidi et al., 2008; Frijns et al., 2016), recent research indicates that decomposing VIX components can enhance forecasting accuracy (Chen et al., 2021). Our research extends this framework by showing that Bitcoin overnight returns enhance VIX prediction accuracy, which in turn improves the performance of VIX derivatives trading strategies. This highlights the broader role of cryptocurrency-based indicators in volatility modeling and risk management.

Third, our study contributes to the growing literature on overnight risk in asset pricing. Building on Taylor's (2007) work on modeling overnight risk, subsequent research has identified distinct patterns in overnight returns. Notable findings include greater cross-sectional dispersion in overnight betas compared to CAPM betas (Bollerslev et al., 2016) and differences between trading and non-trading periods in terms of information flow, liquidity, and borrowing costs (Lou et al., 2019).³ Our analysis extends these insights by

³ Recent work by Bogousslavsky (2021) shows that overnight position holding exposes investors to significant illiquidity and high trading costs. Akbas et al. (2022) find that the frequency of positive overnight returns followed by negative intraday reversals predicts future cross-sectional returns.

showing that Bitcoin overnight returns not only reflect market-wide sentiment but also predict market uncertainty, demonstrating their unique role in bridging cryptocurrency and traditional financial markets.

Finally, our study adds to the literature on the evolving interconnection between cryptocurrency and traditional financial markets. Jain et al. (2019) provide an early comparative analysis of the Bitcoin market relative to traditional asset markets, but this relationship has since evolved substantially. Our findings indicate that Bitcoin's overnight returns enhance VIX prediction and improve the profitability of VIX-related trading strategies, suggesting that cryptocurrencies are becoming increasingly integrated into mainstream financial market dynamics. This contrasts with earlier views of cryptocurrency market isolation (Borri, 2019; Liu and Tsyvinski, 2021) and instead supports the notion that Bitcoin now plays a key role in forecasting volatility in traditional financial markets.

The remainder of this paper is organized as follows. Section 2 examines Bitcoin overnight returns as a proxy for investor sentiment. Section 3 analyzes their predictive power for VIX movements, both in-sample and out-of-sample. Section 4 evaluates the economic significance of the forecasting models through trading strategies. Section 5 presents robustness tests, and Section 6 concludes.

2. Bitcoin Overnight Returns and Investor Sentiment

2.1 Bitcoin Return Decomposition

Our analysis employs five-minute Bitcoin price data from CoinAPI, a leading cryptocurrency research provider (Griffin and Shams, 2020; Oefele et al., 2024), covering the period from January 2018 to December 2023. We align Bitcoin prices with U.S. stock market trading hours (9:30 AM to 4:00 PM Eastern Time, Monday to Friday, excluding market

holidays) and convert prices from Coordinated Universal Time (UTC) to Eastern Time (ET). The period outside trading hours is defined as the overnight session.

After alignment, we decompose daily Bitcoin returns ($R_{BTC,t}^{CC}$) into overnight returns ($R_{BTC,t}^{ON}$) and trading-hour returns ($R_{BTC,t}^{TH}$) as follows:

$$R_{BTC,t}^{CC} = \ln BTC_t^{Close} - \ln BTC_{t-1}^{Close} \quad (1)$$

$$R_{BTC,t}^{ON} = \ln BTC_t^{Open} - \ln BTC_{t-1}^{Close} \quad (2)$$

$$R_{BTC,t}^{TH} = \ln BTC_t^{Close} - \ln BTC_t^{Open} \quad (3)$$

where BTC_t^{Close} represents Bitcoin's closing price and BTC_t^{Open} represents the opening price. By definition, overnight returns occur before the same day's trading hours.

Panel A of Table 1 presents descriptive statistics for daily Bitcoin returns and their temporal components. The results show a pattern of return reversals between overnight and trading hours. Specifically, overnight returns ($R_{BTC,t}^{ON}$) have an average of 0.093%, whereas trading-hour returns ($R_{BTC,t}^{TH}$) show a mean of -0.029%. The mean overnight returns are substantially higher than the trading-hour returns, with an average difference of 0.122%.

< Insert Table 1 Here >

2.2 Identifying Bitcoin Overnight Returns as a Sentiment Measure

We investigate Bitcoin overnight returns as a potential proxy for investor sentiment in cryptocurrency markets, drawing on theoretical frameworks established in traditional financial markets (Berkman et al., 2012; Aboody et al., 2018).⁴ Building on these studies, we hypothesize that if Bitcoin overnight returns reflect retail investor sentiment, they should exhibit positive returns during the overnight period, followed by reversals during the trading

⁴ Berkman et al. (2012) and Aboody et al. (2018) explore the use of overnight returns as a measure of firm-specific investor sentiment in stock markets. They document a strong tendency for positive returns during the overnight period, followed by reversals during trading hours. This pattern is particularly pronounced in stocks that have recently attracted retail investor attention.

day—a pattern referred to as “daily return reversals”, as shown in Panel A of Table 1. In particular, the magnitude of these reversals should increase with investor attention.

2.2.1 Univariate Sorting of Bitcoin Returns Based on Investor Attention

Table 2 presents univariate sorts of Bitcoin return components conditional on abnormal investor attention measures. The sample is sorted into deciles based on abnormal attention levels, with Decile 1 (lowest attention) labeled as “Low,” Decile 10 (highest attention) labeled as “High,” and Deciles 2–9 grouped as “Mid”, based on the previous day’s abnormal investor attention. For each of these three attention-based portfolios, we report the average overnight Bitcoin return ($R_{BTC,t}^{ON}$), the average trading-hour return ($R_{BTC,t}^{TH}$), and the daily return reversal ($R_{BTC,t}^{ON} - R_{BTC,t}^{TH}$), respectively. The “High-Low” row reports the return difference between the high- and low-attention deciles.

We use two proxies to capture retail investors’ attention in the cryptocurrency market. The first proxy is Google search volume for the keyword “Bitcoin.” The second proxy is Bitcoin’s daily unique active addresses, which directly reflect retail trading participation. For both attention measures, we follow Jiang et al. (2022) and compute abnormal investor attention as the deviation of each proxy from its trailing monthly average, normalized by the monthly mean.⁵

< Insert Table 2 Here >

Table 2 shows that overnight returns increase consistently with investor attention, while trading-hour returns display no clear pattern. The difference in overnight returns between low- and high-attention portfolios (High-Low) is both economically and statistically significant for both attention measures: 1.133% (t -statistic = 2.072) for Google Search, and

⁵ We adopt an alternative methodology to classify abnormal investor attention using a time-series partition approach, following Entrop et al. (2020). The results are robust and remain consistent.

0.987% (t -statistic = 2.033) for unique active addresses. In contrast, trading-hour return differentials are not statistically significant, with High-Low spreads of -0.114% (t -statistic = -0.361) for Google Search and -0.288% (t -statistic = -0.880) for unique active addresses. The High-Low differential in daily return reversals ($R_{BTC,t}^{ON} - R_{BTC,t}^{TH}$) is also statistically significant for both attention proxies: 1.247% (t -statistic = 2.037) for Google Search and 1.275% (t -statistic = 2.216) for unique active addresses. These findings suggest that sentiment-driven price dynamics are concentrated in overnight trading. The increase in return reversal magnitude with investor attention provides strong evidence of sentiment-driven trading patterns in the Bitcoin market, where higher levels of abnormal investor attention are associated with larger overnight-to-trading-hour return reversals.

2.2.2 Alternative Investor Attention Metric

Following Lyócsa et al. (2020), we construct an alternative investor attention metric based on a composite Google Search Index for Bitcoin-related terms.⁶ We replicate our portfolio sort analysis using this alternative measure in Appendix A. Overnight Bitcoin returns increase monotonically with investor attention, whereas trading-hour returns show no clear pattern. The High-Low decile spread in overnight returns remains both economically and statistically significant, while trading-hour returns show no significant pattern across attention deciles. Additionally, the High-Low decile differential in daily return reversals remains significant for the composite Google Search: 1.152% (t -statistic = 2.306). These results support our hypothesis that Bitcoin overnight returns capture sentiment-driven overreaction, which subsequently reverses during trading hours. Furthermore, they confirm

⁶ Specifically, we apply Principal Component Analysis (PCA) to aggregate multiple Bitcoin-related search queries, including terms such as “Bitcoin,” “Bitcoin stock,” “Bitcoin hedge,” “Bitcoin S&P 500,” “Bitcoin currency,” “Bitcoin ban,” “Coinbase,” and “Binance.” Our empirical results remain robust across alternative specifications of search terms used in constructing the Google Search Index.

that the relationship between Bitcoin overnight returns and investor attention is robust to alternative attention measures.

Overall, our decomposition analysis indicates that cross-decile variation in Bitcoin returns is primarily driven by overnight price movements, while trading-hour return differentials remain insignificant. The timing of these overnight price formations relative to U.S. market openings, along with their relationship to investor attention, suggests that Bitcoin overnight returns capture sentiment dynamics. These findings motivate further investigation into whether such sentiment signals have predictive value for broader market uncertainty.

3. Forecasting the VIX: The Role of Overnight Bitcoin Returns

Having established Bitcoin overnight returns as a proxy for market sentiment, we now examine their relationship with expected stock market volatility, as measured by the VIX index, which gauges anticipated stock market volatility over a 30-day horizon. Our investigation includes both in-sample and out-of-sample analyses to evaluate the predictive power of Bitcoin overnight returns on VIX dynamics.⁷

3.1 In-Sample Analysis

To assess the incremental predictive power of Bitcoin returns on VIX movements, we examine the relationship between Bitcoin's overnight and trading-hour returns and subsequent intraday VIX changes. While prior research (Harvey and Whaley, 1992; Simon, 2003; Koopman et al., 2005; Becker et al., 2009; Ding et al., 2021) has identified several determinants of trading-hour VIX changes, we extend the literature by investigating the unique role of cryptocurrency markets. Following Chen et al. (2021), we specify a model that

⁷ We calculate daily log changes in the VIX from close to close ($\Delta \ln V_t^{CC}$) and further decompose them into overnight ($\Delta \ln V_t^{ON}$) and trading-hour ($\Delta \ln V_t^{TH}$) components, defined analogously to Bitcoin returns in Equations (1) to (3). Panel B of Table 1 presents descriptive statistics for these VIX changes.

accounts for holiday effects and Federal Open Market Committee (FOMC) announcement impacts:

$$\begin{aligned} \Delta \ln V_t^{TH} = & \alpha + \beta_1 R_{BTC,t}^{ON} + \beta_2 R_{BTC,t-1}^{TH} + \gamma_1 R_{SPF,t}^{ON} + \gamma_2 R_{SPF,t-1}^{TH} + \gamma_3 \Delta \ln V_t^{ON} \\ & + \sum_{i=1}^2 \phi_i NTD_{t+1-i} + \sum_{i=1}^3 \psi_i D_{t+2-i}^{FOMC} + e_t \end{aligned} \quad (4)$$

where $\Delta \ln V_t^{TH}$ denotes trading-hour changes in the VIX index on day t . For SPX futures,⁸ $R_{SPF,t}^{ON}$ represents contemporaneous overnight returns and $R_{SPF,t-1}^{TH}$ represents lagged trading-hour returns. For Bitcoin, $R_{BTC,t}^{ON}$ captures contemporaneous overnight returns and $R_{BTC,t-1}^{TH}$ represents lagged trading-hour returns. Holiday effects are captured by dummy variables NTD_t and NTD_{t-1} , where $NTD_t = 1$ if day t is the first trading day following a non-trading period and 0 otherwise. FOMC announcement effects are captured by D_{t+1}^{FOMC} , D_t^{FOMC} , and D_{t-1}^{FOMC} , where $D_t^{FOMC} = 1$ on pre-scheduled FOMC announcement days and 0 otherwise. The model also controls for contemporaneous overnight VIX changes ($\Delta \ln V_t^{ON}$) and includes three lags of both overnight and trading-hour VIX changes, with the optimal lag length determined by the Akaike Information Criterion (AIC).

< Insert Table 3 Here >

Table 3 reveals several notable patterns in the relationship between Bitcoin returns and subsequent VIX changes. The first two columns present benchmark specifications, showing predictive power for both lagged trading-hour returns from SPX futures and Bitcoin. Columns (1) and (2) show that the coefficient of $R_{SPF,t-1}^{TH}$ is significant at the 1% level. In Column (2), the coefficient of $R_{BTC,t-1}^{TH}$ is significant at the 10% level. Importantly, Column (3) examines the predictive power of Bitcoin overnight returns on subsequent trading-hour VIX changes, controlling for contemporaneous overnight VIX changes and lagged VIX

⁸ Our methodology controls for S&P 500 (SPX) futures returns to isolate Bitcoin trading's unique informational content. This approach is motivated by Bitcoin's 24/7 global trading characteristics (Makarov and Schoar, 2019), which we hypothesize enable it to capture broader sentiment factors than the domestically oriented SPX futures market.

changes. The coefficient of $R_{BTC,t}^{ON}$ is negative and statistically significant (-0.262, t -statistic = -2.49), indicating that increased sentiment in the overnight Bitcoin market predicts lower equity market uncertainty during subsequent trading hours. The economic magnitude is substantial: a one-standard-deviation increase in Bitcoin overnight returns is associated with a 13% decrease in the standard deviation of trading-hour VIX changes.⁹

The predictive relationship remains robust across multivariate specifications. After controlling for SPX futures component returns and holiday effects in Column (4), the coefficient of $R_{BTC,t}^{ON}$ remains significant (-0.255, t -statistic = -2.44), suggesting that Bitcoin overnight returns contain incremental predictive information beyond conventional equity market movements and seasonal patterns. The results hold in Column (5) after incorporating FOMC announcement effects, indicating that Bitcoin overnight returns capture sentiment components distinct from macroeconomic news releases. Notably, when Bitcoin overnight returns are included in Columns (4) and (5), the coefficients on $R_{BTC,t-1}^{TH}$ become statistically insignificant, while the coefficients on $R_{BTC,t}^{ON}$ retain their magnitude and significance.

In summary, the predictive power for VIX changes is concentrated in overnight Bitcoin returns, underscoring the importance of overnight cryptocurrency market activity in forecasting next-day equity market uncertainty.

3.2 Out-of-Sample Analysis

This section evaluates the predictive power of Bitcoin overnight returns on VIX changes using out-of-sample analysis. Our methodology employs two model families: Vector Autoregressive (VAR) and Heterogeneous Autoregressive (HAR) specifications. For all model estimations, we use a rolling window of 500 observations, representing one-third of

⁹ A one-standard-deviation increase in Bitcoin overnight returns leads to a 0.0094 (0.036×0.262) decrease in VIX changes, corresponding to a 13% decrease in the standard deviation of $\Delta \ln V_t^{TH}$ ($= 0.036 \times 0.262 / 0.072$).

the sample period, to estimate model parameters and generate one-day-ahead out-of-sample forecasts.¹⁰ Appendix B provides a comprehensive summary of all model specifications employed in the out-of-sample analysis.

Specifically, the forecasting framework aggregates predicted overnight and trading-hour VIX changes to construct forecasts of the VIX closing value on day t ($\widehat{\ln V_t^C}$):

$$\widehat{\ln V_t^C} = \ln V_{t-1}^C + \Delta \widehat{\ln V_t^{ON}} + \Delta \widehat{\ln V_t^{TH}} \quad (5)$$

where $\ln V_{t-1}^C$ is the log closing level of VIX on the previous trading day, and $\Delta \widehat{\ln V_t^{ON}}$ and $\Delta \widehat{\ln V_t^{TH}}$ represent the forecasted overnight and trading-hour VIX changes, respectively.

We employ three baseline VAR specifications in our empirical analysis to model the dynamics of overnight and trading-hour VIX changes:

$$\Delta \ln V_t^{ON} = \alpha_0 + \beta_1 R_{SPF,t-1}^{TH} + \boldsymbol{\gamma}' \mathbf{X}_t + e_t^{ON} \quad (6)$$

$$\Delta \ln V_t^{TH} = \alpha_0 + \beta_1 R_{SPF,t}^{ON} + \beta_2 R_{SPF,t-1}^{TH} + \beta_3 \Delta \ln V_t^{ON} + \boldsymbol{\gamma}' \mathbf{X}_t + e_t^{TH} \quad (7)$$

where $\mathbf{X}_t$ contains control variables that vary across three baseline specifications. VAR1 includes no additional control variables. VAR2 extends the model by incorporating non-trading day indicators (NTD_t , NTD_{t-1}). VAR3 further extends the model by including FOMC announcement indicators (D_{t+1}^{FOMC} , D_t^{FOMC} , D_{t-1}^{FOMC}). All specifications include three lags of both overnight and trading-hour VIX changes to account for VIX dynamics, with the optimal lag length determined using the AIC criterion.

We also extend our baseline VAR specifications by incorporating Bitcoin returns, generating nine additional models. Specifically, each baseline specification (VAR1, VAR2, and VAR3) is augmented with three distinct Bitcoin return measures: overnight returns (ON), trading-hour returns (TH), and close-to-close returns (CC). For example, the augmentation of

¹⁰ The initial rolling window covers from January 2018 to December 2019.

VAR1 yields VAR1-ON, VAR1-TH, and VAR1-CC. A similar augmentation is applied to the VAR2 and VAR3 specifications.

To complement our VAR analysis, and following Jiang and Qiao (2022), we implement a Heterogeneous Autoregressive model with exogenous variables (HARX) for VIX closing level prediction:

$$\ln V_t^C = \alpha + \beta_1 \ln V_{t-1}^C + \beta_2 \ln V_{t-1}^{C(5)} + \beta_3 \ln V_{t-1}^{C(22)} + \delta_1 R_{SPF,t}^{ON} + \delta_2 R_{SPF,t-1}^{TH} + \mathbf{Y}' \mathbf{Z}_t + e_t^C \quad (8)$$

where $\ln V_t^C$ denotes the logarithm of the VIX closing level, and $\ln V_t^{C(k)}$ represents the k -day moving average of log VIX levels from day $t - 1$ to day $t - k$. The vector $\mathbf{Z}_t$ includes additional predictors that vary across specifications. In the baseline HAR model, $\mathbf{Z}_t$ is an empty vector. The HAR-ON specification sets $\mathbf{Z}_t = R_{BTC,t}^{ON}$, incorporating Bitcoin overnight returns. The HAR-TH specification uses $\mathbf{Z}_t = R_{BTC,t-1}^{TH}$, incorporating Bitcoin trading-hour returns. The HAR-CC specification employs $\mathbf{Z}_t = R_{BTC,t-1}^{CC}$, incorporating Bitcoin close-to-close returns.

We conduct a comprehensive evaluation of the out-of-sample forecasting performance for each model specification. Table 4 presents three standard loss metrics: Mean Squared Error (MSE), Mean Absolute Percentage Error (MAPE), and Mean Absolute Error (MAE), along with the coefficient of determination (R^2) as a measure of model fit. The HAR specification augmented with Bitcoin overnight returns (HAR-ON) demonstrates superior predictive accuracy, achieving the lowest MAPE and MAE among all specifications. Within the VAR framework, the VAR2-ON specification, which incorporates Bitcoin overnight returns and holiday effects, exhibits optimal performance with the lowest MSE and the highest explanatory power. The VAR3-ON specification, which further extends the model by incorporating FOMC announcement effects, maintains comparable predictive efficacy, as evidenced by its high R^2 .

< Insert Table 4 Here >

Table 5 presents the Diebold-Mariano (DM) test statistics evaluating the relative forecasting accuracy between models incorporating Bitcoin overnight returns and alternative specifications, using MAE as the loss function. The null hypothesis posits equivalent forecasting accuracy between competing models. A significantly positive (negative) DM test statistic indicates that the row model has superior (inferior) forecasting performance relative to the column model. The test statistics reveal two key patterns. First, specifications utilizing close-to-close Bitcoin returns exhibit significantly inferior forecasting performance relative to their overnight-return counterparts, as evidenced by negative and statistically significant DM statistics. Second, models excluding Bitcoin returns or incorporating trading-hour Bitcoin returns generally underperform their overnight-return counterparts, although these differences are not statistically significant at conventional levels.

< Insert Table 5 Here >

In Table 6, the Model Confidence Set (MCS) results further reinforce our findings. For each model, p -values are reported for four loss metrics: MSE, heteroskedasticity-adjusted MSE (HMSE), MAE, and MAPE. Notably, both the HAR-ON and VAR2-ON specifications emerge as the top-performing predictive models, achieving maximum p -values ($p=1.000$) across all four loss metrics, underscoring their superior performance.

< Insert Table 6 Here >

Moreover, we assess the sensitivity of our results to the estimation window specification by changing the rolling window from 500 to 750 observations, which represents half of the sample size. Appendix C provides a comprehensive evaluation of out-of-sample forecasting performance under this alternative parameterization. Panel A shows that the HAR-ON specification exhibits superior predictive accuracy, achieving the lowest MSE and the highest R^2 . The VAR3-ON specification similarly demonstrates strong performance, generating the lowest MAPE and MAE. Panel B presents Diebold-Mariano (DM) test statistics, comparing

forecasting accuracy between models incorporating Bitcoin overnight returns and alternative specifications. The significantly negative test statistics indicate the superior forecasting accuracy of models incorporating Bitcoin overnight returns relative to alternative specifications. Panel C reports Model Confidence Set (MCS) results, which further support these findings. Both HAR-ON and VAR3-ON specifications achieve maximum p -values ($p = 1.000$) for at least two evaluation criteria, confirming their superior forecasting performance. Appendix C provides strong evidence for the robustness of our main findings to alternative estimation windows.

Overall, our innovative approach, which synchronizes Bitcoin return decomposition with U.S. equity market hours, yields significant findings. These empirical results confirm Bitcoin overnight returns as a significant predictor of subsequent VIX changes, uncovering distinct information content in overnight price formation. Even after controlling for previously documented determinants of VIX dynamics, our methodological framework demonstrates superior forecasting accuracy.

4. Economic Implication of VIX Predictability

We assess the economic significance of our predictive models by implementing long-short trading strategies. Since the VIX index itself is not directly tradable, we use the ProShares VIX Short-Term Futures ETF (VIXY) as the trading instrument. VIXY tracks the performance of the VIX index, making it a suitable instrument for trading VIX fluctuations.¹¹

Our trading strategies adopt a directional approach, meaning we take long (buy) or short (sell) positions based on the predicted direction of VIX changes rather than their magnitude. We examine two model families (HARX and VARX) by implementing dynamic long-short

¹¹ VIXY provides long exposure to the S&P 500 VIX Short-Term Futures Index, which holds a portfolio of monthly VIX futures contracts with a weighted average time to expiration of one month.

strategies, adjusting our positions based on evolving forecasts.¹² For the HAR model, we use daily rebalancing: at the market close on day $t-1$, we take a \$1 long (short) position in VIXY if the model predicts a positive (negative) VIX change from the close of day $t-1$ to the close of day t . All positions are liquidated before the market close on day t . For the VAR model, we employ a semi-daily rebalancing approach. During overnight periods, we take a \$1 long (short) position at the market close on day $t-1$, conditional on a predicted VIX increase (decrease), and close the position at the market open on day t . During trading hours, we take a \$1 long (short) position at the market open on day t , conditional on a forecasted VIX increase (decrease), and close the position at the market close on day t .

We evaluate strategy performance against a passive buy-and-hold benchmark using two metrics: the annualized information ratio (IR) and the annualized certainty equivalent return (CER). The annualized IR is calculated as the ratio of average annualized abnormal returns (long-short strategy returns in excess of buy-and-hold returns) to their standard deviation. The annualized CER, which incorporates risk aversion within a mean-variance framework, measures the risk-return trade-off. The CER gain represents the difference between the CER of the long-short strategy and that of the passive benchmark.¹³ Higher IR and CER gains indicate superior performance of the long-short strategy relative to the benchmark.¹⁴

Table 7 reports the economic performance of long-short trading strategies based on our VIX predictive models. The performance of each strategy is evaluated using five metrics: average daily abnormal returns, their standard deviation, the annualized IR, and the

¹² Consistent with our previous methodology, we use a rolling window of 500 observations to estimate parameters and generate one-step-ahead forecasts.

¹³ Following Rapach and Zhou (2022), we calculate the annualized certainty equivalent return (CER) as: $CER = \mu - 0.5\gamma\sigma^2$, where μ and σ^2 represent the mean and variance of annualized returns, and γ denotes the relative risk aversion coefficient of a mean-variance investor. We examine two levels of risk aversion: $\gamma = 1$ for low-risk aversion and $\gamma = 3$ for high-risk aversion.

¹⁴ Following Livingston and Zhou (2015), we account for transaction costs of 9.7 basis points per dollar traded.

annualized CER gains for investors with low and high-risk aversion. The results highlight the economic value of incorporating Bitcoin overnight returns into the strategies.

< Insert Table 7 Here >

Specifically, the HAR-ON specification, which enhances the baseline HAR model with Bitcoin overnight returns, demonstrates outstanding performance across both economic metrics. First, the strategy achieves an annualized IR of 0.986, meaning that for each unit of annualized risk (measured by the standard deviation of abnormal returns), the strategy generates 0.986 times the annualized abnormal returns above the passive benchmark. The CER gains further highlight the economic value of the long-short strategy. For investors with low-risk aversion, the strategy yields CER gains of 1.20, providing 1.20 additional units of utility annually compared to the benchmark. Even for investors with high-risk aversion, the strategy yields CER gains of 0.519 per year, demonstrating strong performance despite greater risk sensitivity. Among VAR models with semi-daily rebalancing, the VAR3-ON model exhibits the best performance. This model integrates Bitcoin overnight returns while accounting for holiday and FOMC announcement effects, achieving an IR of 0.406 and CER gains of 0.434 and 0.136 for low- and high-risk aversion investors, respectively.

We test the robustness of our economic implication results by implementing trading strategies with an alternative VIX derivative instrument. Specifically, we replace VIXY with the iPath Series B S&P 500 VIX Short-Term Futures ETN (VXX), while maintaining methodological consistency. Appendix D presents the economic performance metrics for our long-short trading strategies using VXX. The empirical patterns strongly support Table 7. In the daily rebalancing framework, the HAR-ON specification maintains its superior performance, generating an annualized IR of 0.894 and CER gains of 1.048 and 0.362 for low- and high-risk-aversion investors, respectively. Among VAR models, the VAR2-ON model exhibits stronger performance, with an IR of 0.322 and CER gains of 0.258 for

low-risk-aversion investors.

Overall, the economic value assessments offer compelling evidence for the practical significance of incorporating Bitcoin overnight returns in VIX forecasting. Models incorporating these returns yield substantial economic gains, as reflected in consistently positive IRs and CER gains across different risk-aversion specifications. These findings support our statistical analysis by demonstrating that Bitcoin overnight returns' predictive power for VIX translates into economically significant trading profits.

5. Further Analysis

This section presents additional empirical tests to verify the robustness of our main findings. First, we conduct out-of-sample analyses comparing pandemic and non-pandemic subperiods. Second, we examine the predictive power of Bitcoin overnight returns for alternative implied volatility indices in U.S. equity markets. Third, we extend the out-of-sample analyses to international markets.

5.1 Pandemic and Non-Pandemic Subperiods

To assess whether the predictive power of Bitcoin overnight returns remains consistent across different market conditions, we divide the out-of-sample period into two subperiods: the pandemic era (January 1, 2020, to May 5, 2023) and the post-pandemic period (May 6, 2023, to December 31, 2023). This division allows us to determine whether our findings are driven by the unique market dynamics that emerged during the COVID-19 crisis.

Table 8 reports out-of-sample forecasting performance during the pandemic period. Panel A indicates that during this period of unprecedented market volatility, the HAR-ON model demonstrates superior predictive accuracy based on the MAPE and MAE metrics, and the VAR2-ON specification excels in terms of the MSE and R^2 metrics. In Panel B, DM test

results show that models incorporating Bitcoin overnight returns significantly outperform alternative specifications. The MCS results in Panel C reveal that both the HAR-ON and VAR2-ON specifications achieve maximum p -values ($p = 1.000$) across all four loss metrics.

< Insert Table 8 Here >

Table 9 reports out-of-sample forecasting performance during the non-pandemic period. Panel A shows that the HAR-ON specification maintains its predictive advantage under both the MSE and R^2 metrics. The MCS results in Panel C confirm that the HAR-ON model's robust performance extends into the post-pandemic period, again achieving maximum p -values ($p = 1.000$) across all four loss metrics, reinforcing the stability of its predictive power.

< Insert Table 9 Here >

To summarize, our subsample analysis suggests that Bitcoin overnight returns possess distinctive predictive power for the VIX across both pandemic and non-pandemic periods, providing consistent signals under varying market conditions.

5.2 Alternative Implied Volatility Indices for U.S. Markets

To further validate the predictive power of Bitcoin overnight returns for U.S. equity markets, we examine two additional CBOE implied volatility indices: the CBOE NASDAQ Volatility Index (VXN)¹⁵, derived from NASDAQ-100 Index options, and the CBOE DJIA Volatility Index (VXD)¹⁶, derived from options on the Dow Jones Industrial Average.

Table 10 presents the forecasting results for these two alternative CBOE implied volatility indices. For VXN forecasting (Panel A), the VAR2-ON model performs best under

¹⁵ The CBOE NASDAQ-100 Volatility Index (VXN) measures the expected 30-day volatility based on NASDAQ-100 Index option prices.

¹⁶ The CBOE DJIA Volatility Index (VXD) is a VIX-style estimate of the expected 30-day volatility in Dow Jones Industrial Average (DJIA) returns.

the MSE and R^2 metrics, and the HAR-ON model excels under the MAPE and MAE criteria. For VXD forecasting (Panel B), the HAR-ON model demonstrates superior performance across all evaluation metrics. These consistent results confirm that the unique predictive power of Bitcoin overnight returns for implied volatility extends robustly across different U.S. equity market indices, suggesting broader implications for U.S. equity market volatility forecasting.

< Insert Table 10 Here >

5.3 International Evidence

We extend our analysis to evaluate the predictive power of Bitcoin overnight returns in international equity markets. To capture both emerging and developed markets, we examine two implied volatility indices: the CBOE Emerging Markets Volatility Index (VXEEM)¹⁷ and the CBOE EFA ETF Volatility Index (VXEFA)¹⁸. Both indices are approximately calculated and disseminated during U.S. market hours, enabling direct comparison with our primary analyses.

Table 11 summarizes the forecasting accuracy for these international volatility indices. For VXEEM forecasting (Panel A), the HAR-ON model demonstrates superior performance across all evaluation criteria. For VXEFA forecasting (Panel B), the HAR-ON model achieves optimal performance under MSE, MAE, and R^2 metrics. These findings align with our main results, confirming that the predictive power of Bitcoin overnight returns extends beyond the U.S. equity market to both emerging and developed international markets. This

¹⁷ The CBOE Emerging Markets ETF Volatility Index (VXEEM) measures the expected 30-day volatility of the iShares MSCI Emerging Markets ETF (EEM), which tracks large- and mid-capitalization stocks in emerging markets.

¹⁸ The CBOE EFA ETF Volatility Index (VXEFA) measures the expected 30-day volatility of the iShares MSCI EAFE ETF (EFA), which tracks stocks in developed markets across Europe, Australia, Asia, and the Far East.

extension suggests that Bitcoin overnight returns capture global sentiment factors that influence implied volatility across diverse market environments.

< Insert Table 11 Here >

6. Conclusion

This study examines the informational content of Bitcoin returns through a novel decomposition framework aligned with U.S. stock market trading hours. Our analysis shows that Bitcoin's overnight returns capture a distinct dimension of investor sentiment, which significantly improves the predictability of equity market volatility dynamics. Specifically, by using Bitcoin's unique active addresses and Google search volume as proxies for retail investor attention, we document that the magnitude of overnight-to-trading-hour return reversals increases with investor attention levels, reinforcing the role of cryptocurrency markets as a barometer of investor sentiment.

We next explore the predictive power of Bitcoin's overnight returns for VIX dynamics. In-sample analysis shows a significantly negative relationship between Bitcoin's overnight returns and subsequent trading-hour VIX changes. Out-of-sample analysis demonstrates that models incorporating Bitcoin's overnight returns outperform baseline models across various evaluation metrics. This predictability remains statistically and economically significant, even when accounting for established determinants of VIX dynamics. The economic significance of these findings is further validated through long-short trading strategies in VIX derivatives, where models incorporating Bitcoin's overnight returns generate substantial gains across different risk preferences.

Our findings have important implications for both market participants and regulators. For investors, Bitcoin's overnight returns serve as a novel sentiment indicator that can enhance volatility forecasting and risk management. The observed predictability suggests that

existing volatility models could benefit from the inclusion of cryptocurrency market signals. From a regulatory perspective, our results highlight the growing interconnection between cryptocurrency and traditional financial markets, emphasizing the need for coordinated oversight in monitoring potential spillover effects.

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Table 1 Descriptive Statistics

This table provides summary statistics for daily Bitcoin returns and VIX index changes from January 1, 2018 to December 31, 2023. Panel A reports Bitcoin returns ($R_{BTC,t}^{CC}$) and their temporal components. Daily Bitcoin returns are decomposed into overnight returns ($R_{BTC,t}^{ON}$), measured from market close to the next-day market open, and trading-hour returns ($R_{BTC,t}^{TH}$), measured from market open to market close. Panel B similarly decomposes daily VIX index changes and reports the statistics for close-to-close changes ($\Delta \ln V_t^{CC}$), overnight changes ($\Delta \ln V_t^{ON}$), and trading-hour changes ($\Delta \ln V_t^{TH}$).

Panel A Bitcoin Return Components			
	$R_{BTC,t}^{CC}$	$R_{BTC,t}^{ON}$	$R_{BTC,t}^{TH}$
Mean (%)	0.064	0.093	-0.029
Std. Dev	0.043	0.036	0.022
Obs.	1512	1512	1512

Panel B VIX Index Changes			
	$\Delta \ln V_t^{CC}$	$\Delta \ln V_t^{ON}$	$\Delta \ln V_t^{TH}$
Mean (%)	0.016	0.964	-0.955
Std. Dev	0.081	0.038	0.072
Obs.	1508	1508	1508

Table 2 Univariate Sorting of Bitcoin Returns Based on Investor Attention

This table reports Bitcoin return components for decile portfolios sorted by abnormal investor attention measures. The sample is first sorted into “Low” (lowest attention decile), “High” (highest attention decile), and “Mid” (intermediate deciles) based on the previous day’s abnormal investor attention. For each portfolio, we report the average overnight returns ($R_{BTC,t}^{ON}$), trading-hour returns ($R_{BTC,t}^{TH}$), and the difference between overnight and trading-hour returns ($R_{BTC,t}^{ON} - R_{BTC,t}^{TH}$). The High-Low row presents the return difference between the high- and low-attention deciles, with the corresponding t -statistics in parentheses. Panel A sorts returns by Google search volume for the keyword “Bitcoin,” where abnormal attention is measured as the deviation from the previous month’s average search volume, normalized by the monthly mean. Panel B sorts returns by Bitcoin’s unique active addresses, where abnormal attention is similarly measured as the normalized deviation from the previous month’s average number of unique active addresses. The sample period spans from January 2018 to December 2023. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

<i>Abnormal investor attention</i>	$R_{BTC,t}^{ON}$	$R_{BTC,t}^{TH}$	$R_{BTC,t}^{ON} - R_{BTC,t}^{TH}$
Panel A: Google search volume of ‘Bitcoin’			
Low (%)	-0.452	-0.166	-0.285
Mid (%)	0.087	0.020	0.067
High (%)	0.681	-0.280	0.962
High - Low (%)	1.133**	-0.114	1.247**
t -statistic	(2.072)	(-0.361)	(2.037)
Panel B: Bitcoin’s unique active addresses			
Low (%)	-0.482	0.136	-0.618
Mid (%)	0.113	-0.034	0.147
High (%)	0.505	-0.152	0.657
High - Low (%)	0.987**	-0.288	1.275**
t -statistic	(2.033)	(-0.880)	(2.216)

Table 3 Predictive Power of Bitcoin Returns for VIX: In-Sample Evidence

This table presents estimates from the following predictive regression, which examines the relationship between Bitcoin overnight returns and trading-hour VIX changes:

$$\Delta \ln V_t^{TH} = \alpha + \beta_1 R_{BTC,t}^{ON} + \beta_2 R_{BTC,t-1}^{TH} + \gamma_1 R_{SPF,t}^{ON} + \gamma_2 R_{SPF,t-1}^{TH} + \gamma_3 \Delta \ln V_t^{ON} + \sum_{i=1}^2 \phi_i NTD_{t+1-i} + \sum_{i=1}^3 \psi_i D_{t+2-i}^{FOMC} + e_t$$

where $\Delta \ln V_t^{TH}$ and $\Delta \ln V_t^{ON}$ represent the trading-hour and overnight changes in the VIX index on day t , respectively. $R_{BTC,t}^{ON}$ and $R_{BTC,t-1}^{TH}$ denote Bitcoin's overnight returns on day t and trading-hour returns on day $t-1$. $R_{SPF,t}^{ON}$ and $R_{SPF,t-1}^{TH}$ represent the SPX futures' overnight returns on day t and trading-hour returns on day $t-1$. The dummy variables NTD_t and NTD_{t-1} capture holiday effects, where $NTD_t=1$ if day t is the first trading day following a non-trading period and 0 otherwise. The dummy variables D_{t+1}^{FOMC} , D_t^{FOMC} , and D_{t-1}^{FOMC} account for FOMC announcement effects, where $D_t^{FOMC}=1$ on pre-scheduled FOMC announcement days and 0 otherwise. All specifications include three lags of both overnight and trading-hour VIX changes to account for VIX dynamics. The sample period spans from January 2018 to December 2023. The reported t -statistics in parentheses are based on Newey-West standard errors. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
$R_{BTC,t}^{ON}$			-0.262** (-2.49)	-0.255** (-2.44)	-0.256** (-2.47)
$R_{BTC,t-1}^{TH}$		-0.140* (-1.70)		-0.099 (-1.18)	-0.113 (-1.36)
$R_{SPF,t}^{ON}$	0.415 (0.51)	0.541 (0.66)		0.504 (0.63)	0.520 (0.65)
$R_{SPF,t-1}^{TH}$	0.841*** (2.61)	0.885*** (2.80)		0.830*** (2.76)	0.849*** (2.85)
$\Delta \ln V_t^{ON}$	-0.054 (-0.82)	-0.040 (-0.60)	-0.083 (-1.27)	-0.068 (-1.04)	-0.062 (-0.93)
NTD_t	0.001 (0.26)	0.003 (0.61)		0.002 (0.48)	0.004 (0.87)
NTD_{t-1}	0.000 (0.00)	0.003 (0.49)		0.000 (0.07)	0.004 (0.62)
D_{t+1}^{FOMC}		-0.010 (-0.92)			-0.011 (-1.02)
D_t^{FOMC}		0.012 (0.87)			0.014 (1.02)
D_{t-1}^{FOMC}		0.033** (2.44)			0.032** (2.42)
<i>Lags</i>	Yes	Yes	Yes	Yes	Yes
Constant	-0.010*** (-3.92)	-0.012*** (-4.76)	-0.009*** (-4.27)	-0.010*** (-3.93)	-0.012*** (-4.82)
Obs.	1499	1498	1499	1499	1498
Adj. R^2	0.010	0.015	0.018	0.026	0.031

Table 4 Out-of-Sample Test: Average Forecasting Losses

This table presents the out-of-sample forecasting performance of various predictive models for the VIX closing level. Model parameters are estimated using a rolling window that covers one-third of the sample size. The table reports average forecasting losses based on four standard evaluation metrics: Mean Squared Error (MSE), Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), and the coefficient of determination (R^2). To ensure comparability, MSE and MAPE values are multiplied by 100, and MAE values are multiplied by 10. Models showing superior performance are highlighted in grey.

	<i>MSE</i>	<i>MAPE</i>	<i>MAE</i>	<i>R</i> ²
<i>HAR</i>	0.644	1.752	0.549	0.940
<i>HAR-ON</i>	0.633	1.737	0.543	0.941
<i>HAR-TH</i>	0.647	1.753	0.549	0.939
<i>HAR-CC</i>	0.647	1.760	0.551	0.939
<i>VAR1</i>	0.634	1.778	0.557	0.941
<i>VAR1-ON</i>	0.625	1.773	0.555	0.941
<i>VAR1-TH</i>	0.635	1.775	0.556	0.940
<i>VAR1-CC</i>	0.636	1.783	0.559	0.940
<i>VAR2</i>	0.629	1.757	0.551	0.941
<i>VAR2-ON</i>	0.620	1.746	0.547	0.942
<i>VAR2-TH</i>	0.631	1.758	0.551	0.941
<i>VAR2-CC</i>	0.632	1.765	0.553	0.941
<i>VAR3</i>	0.633	1.761	0.552	0.941
<i>VAR3-ON</i>	0.623	1.752	0.549	0.942
<i>VAR3-TH</i>	0.635	1.761	0.552	0.940
<i>VAR3-CC</i>	0.636	1.770	0.555	0.940

Table 5 Out-of-Sample Test: Diebold-Mariano Test

This table presents Diebold-Mariano (DM) test statistics, comparing the forecasting accuracy of competing models using MAE as the loss function. The null hypothesis states that the forecasting accuracy of the model in each row is equal to that of the corresponding model in each column. A significantly positive (negative) DM test statistic indicates that the row model has superior (inferior) forecasting performance relative to the column model. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	<i>HAR-ON</i>	<i>VAR1-ON</i>	<i>VAR2-ON</i>	<i>VAR3-ON</i>
<i>HAR</i>	-1.51			
<i>HAR-TH</i>	-1.54			
<i>HAR-CC</i>	-2.13**			
<i>VAR1</i>		-0.90		
<i>VAR1-TH</i>		-0.51		
<i>VAR1-CC</i>		-1.67*		
<i>VAR2</i>			-1.53	
<i>VAR2-TH</i>			-1.60	
<i>VAR2-CC</i>			-2.46**	
<i>VAR3</i>				-1.46
<i>VAR3-TH</i>				-1.38
<i>VAR3-CC</i>				-2.46**

Table 6 Out-of-Sample Test: Model Confidence Set Test

This table presents the results of the Model Confidence Set (MCS) test. For each model, p -values are provided for four loss metrics: MSE, heteroskedasticity-adjusted MSE (HMSE), MAE, and MAPE. Models with p -values marked by * are included in the 90% confidence set, and those marked by ** are included in the 75% confidence set. Models showing superior performance are highlighted in grey.

	<i>MSE</i>	<i>HMSE</i>	<i>MAE</i>	<i>MAPE</i>
<i>HAR</i>	0.796**	0.563**	0.813**	0.889**
<i>HAR-ON</i>	1.000**	1.000**	1.000**	1.000**
<i>HAR-TH</i>	0.494**	0.318**	0.798**	0.854**
<i>HAR-CC</i>	0.353**	0.197*	0.006	0.005
<i>VAR1</i>	0.608**	0.653**	0.613**	0.652**
<i>VAR1-ON</i>	0.995**	0.991**	0.683**	0.657**
<i>VAR1-TH</i>	0.493**	0.568**	0.689**	0.716**
<i>VAR1-CC</i>	0.207*	0.178*	0.259**	0.240*
<i>VAR2</i>	0.513**	0.518**	0.810**	0.904**
<i>VAR2-ON</i>	1.000**	1.000**	1.000**	1.000**
<i>VAR2-TH</i>	0.268**	0.318**	0.791**	0.852**
<i>VAR2-CC</i>	0.124*	0.136*	0.031	0.026
<i>VAR3</i>	0.192*	0.233*	0.444**	0.597**
<i>VAR3-ON</i>	0.894**	0.921**	0.971**	0.979**
<i>VAR3-TH</i>	0.110*	0.145*	0.508**	0.598**
<i>VAR3-CC</i>	0.044	0.060	0.010	0.007

Table 7 Performance of VIXY-Based Trading Strategies

This table evaluates the economic performance of long-short trading strategies using VIXY over the sample period from January 2018 to December 2023. Model parameters are estimated using a rolling window covering one-third of the sample. Performance is assessed using multiple metrics: average daily abnormal returns (in percentage), the standard deviation of daily abnormal returns, the annualized Information Ratio (IR), and annualized certainty equivalent return (CER) gains relative to a passive buy-and-hold strategy. The CER is computed as $CER = \mu - 0.5\gamma\sigma^2$, where μ and σ^2 represent the mean and variance of annualized asset returns, and γ denotes the relative risk aversion coefficient of a mean-variance investor. We consider $\gamma = 1$ for low-risk version and $\gamma = 3$ for high-risk aversion. CER gains measure the difference between the CER of the long-short strategy and that of the passive buy-and-hold strategy. Models showing superior performance are highlighted in grey.

	Average daily abnormal return (%)	Std of daily abnormal return	Annualized I.R.	Annualized CER gains ($\gamma=1$)	Annualized CER gains ($\gamma=3$)
<i>HAR</i>	0.276	0.098	0.459	0.338	-0.455
<i>HAR-ON</i>	0.579	0.096	0.986	1.200	0.519
<i>HAR-TH</i>	0.301	0.098	0.501	0.404	-0.386
<i>HAR-CC</i>	0.221	0.098	0.366	0.181	-0.630
<i>VAR1</i>	0.020	0.096	0.034	-0.279	-0.942
<i>VAR1-ON</i>	0.032	0.095	0.055	-0.231	-0.862
<i>VAR1-TH</i>	-0.018	0.096	-0.032	-0.391	-1.075
<i>VAR1-CC</i>	-0.025	0.095	-0.043	-0.391	-1.037
<i>VAR2</i>	0.161	0.091	0.289	0.205	-0.245
<i>VAR2-ON</i>	0.210	0.090	0.381	0.359	-0.043
<i>VAR2-TH</i>	0.178	0.092	0.317	0.237	-0.239
<i>VAR2-CC</i>	0.172	0.091	0.308	0.237	-0.202
<i>VAR3</i>	0.119	0.091	0.213	0.096	-0.343
<i>VAR3-ON</i>	0.219	0.088	0.406	0.434	0.136
<i>VAR3-TH</i>	0.185	0.093	0.324	0.221	-0.317
<i>VAR3-CC</i>	0.154	0.093	0.271	0.149	-0.372

Table 8 Out-of-Sample Forecasting Performance during the Pandemic Period

This table evaluates the robustness of out-of-sample forecasting performance for 16 predictive models of VIX closing levels during the pandemic period (January 1, 2020, to May 5, 2023). Models exhibiting superior performance are highlighted in grey. Panel A reports average forecasting losses based on four standard evaluation metrics: Mean Squared Error (MSE), Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), and the coefficient of determination (R^2). To ensure comparability, MSE and MAPE values are multiplied by 100, and MAE values are multiplied by 10. Panel B presents Diebold-Mariano (DM) test statistics, comparing the forecasting accuracy of competing models using MAE as the loss function. The null hypothesis states that the forecasting accuracy of the model in each row is equal to that of the corresponding model in each column. A significantly positive (negative) DM test statistic indicates that the row model has superior (inferior) forecasting performance relative to the column model. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. Panel C presents the results of the Model Confidence Set (MCS) test. For each model, p -values are provided for four loss metrics: MSE, heteroskedasticity-adjusted MSE (HMSE), MAE, and MAPE. Models with p -values marked by * are included in the 90% confidence set, and those marked by ** are included in the 75% confidence set.

Panel A: Out-of-sample forecasting performance				
	MSE	MAPE	MAE	R^2
HAR	0.716	1.796	0.574	0.933
HAR-ON	0.703	1.777	0.568	0.934
HAR-TH	0.719	1.797	0.574	0.933
HAR-CC	0.718	1.803	0.577	0.933
VAR1	0.700	1.828	0.584	0.934
VAR1-ON	0.690	1.818	0.581	0.935
VAR1-TH	0.702	1.825	0.583	0.934
VAR1-CC	0.703	1.834	0.586	0.934
VAR2	0.692	1.805	0.577	0.935
VAR2-ON	0.681	1.790	0.572	0.936
VAR2-TH	0.695	1.805	0.577	0.935
VAR2-CC	0.695	1.813	0.580	0.935
VAR3	0.697	1.811	0.579	0.935
VAR3-ON	0.685	1.797	0.574	0.936
VAR3-TH	0.699	1.810	0.579	0.934
VAR3-CC	0.701	1.820	0.582	0.934
Panel B: Diebold-Mariano Test Results				
	HAR-ON	VAR1-ON	VAR2-ON	VAR3-ON
HAR	-2.33**			
HAR-TH	-2.52**			
HAR-CC	-2.57**			
VAR1		-2.49**		
VAR1-TH		-2.47**		
VAR1-CC		-2.81***		
VAR2			-2.96***	
VAR2-TH			-2.99***	
VAR2-CC			-3.40***	

<i>VAR3</i>	-3.00 ^{***}
<i>VAR3-TH</i>	-2.91 ^{***}
<i>VAR3-CC</i>	-3.39 ^{***}

Panel C: Model Confidence Set (MCS)

	<i>MSE</i>	<i>HMSE</i>	<i>MAE</i>	<i>MAPE</i>
<i>HAR</i>	0.874 ^{**}	0.694 ^{**}	0.797 ^{**}	0.856 ^{**}
<i>HAR-ON</i>	1.000 ^{**}	1.000 ^{**}	1.000 ^{**}	1.000 ^{**}
<i>HAR-TH</i>	0.597 ^{**}	0.435 ^{**}	0.788 ^{**}	0.822 ^{**}
<i>HAR-CC</i>	0.555 ^{**}	0.424 ^{**}	0.044	0.051
<i>VAR1</i>	0.510 ^{**}	0.485 ^{**}	0.561 ^{**}	0.647 ^{**}
<i>VAR1-ON</i>	0.928 ^{**}	0.882 ^{**}	0.734 ^{**}	0.636 ^{**}
<i>VAR1-TH</i>	0.371 ^{**}	0.375 ^{**}	0.634 ^{**}	0.709 ^{**}
<i>VAR1-CC</i>	0.219 [*]	0.234 [*]	0.363 ^{**}	0.381 ^{**}
<i>VAR2</i>	0.471 ^{**}	0.411 ^{**}	0.700 ^{**}	0.790 ^{**}
<i>VAR2-ON</i>	1.000 ^{**}	1.000 ^{**}	1.000 ^{**}	1.000 ^{**}
<i>VAR2-TH</i>	0.237 [*]	0.245 [*]	0.728 ^{**}	0.769 ^{**}
<i>VAR2-CC</i>	0.106 [*]	0.101 [*]	0.108 [*]	0.109 [*]
<i>VAR3</i>	0.146 [*]	0.146 [*]	0.232 [*]	0.302 ^{**}
<i>VAR3-ON</i>	0.874 ^{**}	0.907 ^{**}	0.870 ^{**}	0.871 ^{**}
<i>VAR3-TH</i>	0.080	0.089	0.311 ^{**}	0.332 ^{**}
<i>VAR3-CC</i>	0.028	0.031	0.026	0.038

Table 9 Out-of-Sample Forecasting Performance during the Non-Pandemic Period

This table evaluates the robustness of out-of-sample forecasting performance for 16 predictive models of VIX closing levels during the non-pandemic period (May 6, 2023, to December 31, 2023). Models exhibiting superior performance are highlighted in grey. Panel A reports average forecasting losses based on four standard evaluation metrics: Mean Squared Error (MSE), Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), and the coefficient of determination (R^2). To ensure comparability, MSE and MAPE values are multiplied by 100, and MAE values are multiplied by 10. Panel B presents Diebold-Mariano (DM) test statistics, comparing the forecasting accuracy of competing models using MAE as the loss function. The null hypothesis states that the forecasting accuracy of the model in each row is equal to that of the corresponding model in each column. A significantly positive (negative) DM test statistic indicates that the row model has superior (inferior) forecasting performance relative to the column model. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. Panel C presents the results of the Model Confidence Set (MCS) test. For each model, p -values are provided for four loss metrics: MSE, heteroskedasticity-adjusted MSE (HMSE), MAE, and MAPE. Models with p -values marked by * are included in the 90% confidence set, and those marked by ** are included in the 75% confidence set.

Panel A: Out-of-sample forecasting performance				
	MSE	MAPE	MAE	R^2
HAR	0.279	1.528	0.418	0.974
HAR-ON	0.273	1.532	0.418	0.974
HAR-TH	0.280	1.530	0.418	0.974
HAR-CC	0.282	1.538	0.421	0.974
VAR1	0.293	1.519	0.417	0.973
VAR1-ON	0.293	1.541	0.423	0.973
VAR1-TH	0.293	1.518	0.417	0.973
VAR1-CC	0.295	1.523	0.418	0.972
VAR2	0.306	1.511	0.417	0.971
VAR2-ON	0.308	1.524	0.420	0.971
VAR2-TH	0.306	1.515	0.418	0.971
VAR2-CC	0.307	1.520	0.419	0.971
VAR3	0.306	1.508	0.416	0.971
VAR3-ON	0.309	1.522	0.420	0.971
VAR3-TH	0.306	1.509	0.417	0.971
VAR3-CC	0.307	1.517	0.419	0.971
Panel B: Diebold-Mariano Test Results				
	HAR-ON	VAR1-ON	VAR2-ON	VAR3-ON
HAR	-0.59			
HAR-TH	-0.66			
HAR-CC	-0.31			
VAR1		0.20		
VAR1-TH		0.26		
VAR1-CC		0.95		
VAR2			0.60	
VAR2-TH			0.64	
VAR2-CC			1.02	

<i>VAR3</i>	0.52
<i>VAR3-TH</i>	0.60
<i>VAR3-CC</i>	0.89

Panel C: Model Confidence Set (MCS)

	<i>MSE</i>	<i>HMSE</i>	<i>MAE</i>	<i>MAPE</i>
<i>HAR</i>	0.909 ^{**}	0.953 ^{**}	1.000 ^{**}	1.000 ^{**}
<i>HAR-ON</i>	1.000 ^{**}	1.000 ^{**}	1.000 ^{**}	1.000 ^{**}
<i>HAR-TH</i>	0.878 ^{**}	0.932 ^{**}	1.000 ^{**}	1.000 ^{**}
<i>HAR-CC</i>	0.016	0.040	0.020	0.121 [*]
<i>VAR1</i>	0.888 ^{**}	0.973 ^{**}	1.000 ^{**}	1.000 ^{**}
<i>VAR1-ON</i>	0.868 ^{**}	0.958 ^{**}	0.994 ^{**}	0.995 ^{**}
<i>VAR1-TH</i>	0.887 ^{**}	0.972 ^{**}	1.000 ^{**}	1.000 ^{**}
<i>VAR1-CC</i>	0.344 ^{**}	0.411 ^{**}	0.912 ^{**}	0.943 ^{**}
<i>VAR2</i>	0.973 ^{**}	0.996 ^{**}	1.000 ^{**}	1.000 ^{**}
<i>VAR2-ON</i>	0.973 ^{**}	0.995 ^{**}	1.000 ^{**}	1.000 ^{**}
<i>VAR2-TH</i>	0.973 ^{**}	0.996 ^{**}	0.860 ^{**}	0.976 ^{**}
<i>VAR2-CC</i>	0.532 ^{**}	0.598 ^{**}	0.184 [*]	0.143 [*]
<i>VAR3</i>	0.972 ^{**}	0.995 ^{**}	1.000 ^{**}	1.000 ^{**}
<i>VAR3-ON</i>	0.968 ^{**}	0.992 ^{**}	1.000 ^{**}	1.000 ^{**}
<i>VAR3-TH</i>	0.973 ^{**}	0.996 ^{**}	0.999 ^{**}	1.000 ^{**}
<i>VAR3-CC</i>	0.772 ^{**}	0.787 ^{**}	0.173 [*]	0.218 [*]

Table 10 Alternative Implied Volatility Indices for U.S. Markets

This table presents the out-of-sample forecasting performance of various predictive models for the closing level of alternative implied volatility indices in U.S. markets. Forecasting performance is evaluated based on four standard evaluation metrics: Mean Squared Error (MSE), Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), and the coefficient of determination (R^2). To ensure comparability, MSE and MAPE values are multiplied by 100, and MAE values are multiplied by 10. Models showing superior performance are highlighted in grey. Panel A reports the average forecasting losses for the CBOE NASDAQ Volatility Index (VXN). Panel B reports the average forecasting losses for the CBOE DJIA Volatility Index (VXD).

Panel A: CBOE NASDAQ Volatility Index				
	MSE	MAPE	MAE	R^2
HAR	0.422	1.326	0.441	0.947
HAR-ON	0.417	1.320	0.438	0.948
HAR-TH	0.424	1.328	0.441	0.947
HAR-CC	0.424	1.332	0.443	0.947
VAR1	0.415	1.359	0.451	0.948
VAR1-ON	0.409	1.364	0.452	0.949
VAR1-TH	0.416	1.360	0.451	0.948
VAR1-CC	0.418	1.363	0.453	0.948
VAR2	0.411	1.346	0.447	0.949
VAR2-ON	0.405	1.346	0.447	0.949
VAR2-TH	0.413	1.347	0.447	0.948
VAR2-CC	0.414	1.353	0.450	0.948
VAR3	0.414	1.353	0.450	0.948
VAR3-ON	0.408	1.354	0.449	0.949
VAR3-TH	0.416	1.355	0.450	0.948
VAR3-CC	0.417	1.361	0.452	0.948
Panel B: CBOE DJIA Volatility Index				
HAR	1.340	2.297	0.648	0.866
HAR-ON	1.332	2.281	0.643	0.867
HAR-TH	1.344	2.298	0.648	0.866
HAR-CC	1.344	2.302	0.649	0.866
VAR1	1.577	2.491	0.706	0.843
VAR1-ON	1.577	2.491	0.706	0.843
VAR1-TH	1.580	2.490	0.706	0.842
VAR1-CC	1.581	2.492	0.706	0.842
VAR2	1.585	2.479	0.703	0.842
VAR2-ON	1.583	2.478	0.702	0.842
VAR2-TH	1.588	2.479	0.703	0.842
VAR2-CC	1.589	2.484	0.704	0.841
VAR3	1.596	2.497	0.708	0.841
VAR3-ON	1.593	2.495	0.707	0.841
VAR3-TH	1.599	2.497	0.708	0.840
VAR3-CC	1.600	2.503	0.710	0.840

Table 11 Out-of-Sample Test: International Evidence

This table presents the out-of-sample forecasting performance of various predictive models for implied volatility indices in international markets. Forecasting performance is evaluated based on four standard evaluation metrics: Mean Squared Error (MSE), Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), and the coefficient of determination (R^2). To ensure comparability, MSE and MAPE values are multiplied by 100, and MAE values are multiplied by 10. Models showing superior performance are highlighted in grey. Panel A reports the average forecasting losses for the CBOE Emerging Markets Volatility Index (VXEEM). Panel B reports the average forecasting losses for the CBOE EFA ETF Volatility Index (VXEFA).

Panel A: CBOE Emerging Markets Volatility Index				
	<i>MSE</i>	<i>MAPE</i>	<i>MAE</i>	R^2
<i>HAR</i>	0.999	1.942	0.624	0.834
<i>HAR-ON</i>	0.997	1.942	0.623	0.835
<i>HAR-TH</i>	1.007	1.948	0.625	0.833
<i>HAR-CC</i>	1.001	1.944	0.624	0.834
<i>VAR1</i>	1.843	2.892	0.915	0.694
<i>VAR1-ON</i>	1.851	2.899	0.917	0.693
<i>VAR1-TH</i>	1.847	2.893	0.916	0.694
<i>VAR1-CC</i>	1.844	2.893	0.916	0.694
<i>VAR2</i>	1.839	2.875	0.910	0.695
<i>VAR2-ON</i>	1.847	2.886	0.913	0.694
<i>VAR2-TH</i>	1.845	2.876	0.910	0.694
<i>VAR2-CC</i>	1.841	2.877	0.910	0.695
<i>VAR3</i>	1.855	2.889	0.914	0.692
<i>VAR3-ON</i>	1.863	2.900	0.917	0.691
<i>VAR3-TH</i>	1.861	2.891	0.914	0.691
<i>VAR3-CC</i>	1.856	2.890	0.914	0.692
Panel B: CBOE EFA ETF Volatility Index				
<i>HAR</i>	1.489	2.554	0.782	0.869
<i>HAR-ON</i>	1.483	2.553	0.781	0.869
<i>HAR-TH</i>	1.493	2.552	0.782	0.868
<i>HAR-CC</i>	1.495	2.559	0.784	0.868
<i>VAR1</i>	1.749	2.945	0.894	0.846
<i>VAR1-ON</i>	1.747	2.953	0.896	0.846
<i>VAR1-TH</i>	1.744	2.936	0.892	0.846
<i>VAR1-CC</i>	1.756	2.950	0.896	0.845
<i>VAR2</i>	1.740	2.955	0.897	0.846
<i>VAR2-ON</i>	1.740	2.965	0.900	0.846
<i>VAR2-TH</i>	1.736	2.947	0.895	0.847
<i>VAR2-CC</i>	1.748	2.960	0.899	0.846
<i>VAR3</i>	1.753	2.976	0.903	0.845
<i>VAR3-ON</i>	1.752	2.984	0.905	0.845
<i>VAR3-TH</i>	1.749	2.971	0.902	0.846
<i>VAR3-CC</i>	1.761	2.980	0.904	0.845

Appendix A. Univariate Sorting of Bitcoin Returns Based on Alternative Investor Attention

This table reports Bitcoin return components for decile portfolios sorted by an alternative investor attention measure. This metric is constructed using a composite Google Search Index for Bitcoin-related terms, where abnormal attention is measured as the normalized deviation of search volume from its trailing weekly average. The sample is grouped into “Low” (lowest attention decile), “High” (highest decile), and “Mid” (intermediate deciles) based on the previous day’s abnormal investor attention. For each portfolio, we report the average overnight returns ($R_{BTC,t}^{ON}$), trading-hour returns ($R_{BTC,t}^{TH}$), and the difference between overnight and trading-hour returns ($R_{BTC,t}^{ON} - R_{BTC,t}^{TH}$). The High-Low row presents the return difference between the high- and low-attention deciles, with the corresponding t -statistics in parentheses. The sample period spans from January 2018 to December 2023. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

<i>Abnormal investor attention</i>	$R_{BTC,t}^{ON}$	$R_{BTC,t}^{TH}$	$R_{BTC,t}^{ON} - R_{BTC,t}^{TH}$
Low (%)	-0.288	0.047	-0.334
Mid (%)	0.072	-0.019	0.091
High (%)	0.636	-0.182	0.818
High - Low (%)	0.923**	-0.229	1.152**
t -statistic	(2.110)	(-0.810)	(2.306)

Appendix B. Out-of-Sample Models Specifications

This appendix presents the specifications of the models used in out-of-sample analysis. We examine two model families: VARX and HAR. For each family, we first present the baseline model, followed by its extensions that incorporate Bitcoin returns.

B.1 VARX Model Family

All VARX models forecast the VIX closing level as follows:

$$\widehat{\ln V_t^C} = \ln V_{t-1}^C + \Delta \ln V_t^{ON} + \Delta \ln V_t^{TH},$$

where $\Delta \ln V_t^{ON}$ and $\Delta \ln V_t^{TH}$ represent the overnight and trading-hour changes in VIX, respectively, modeled according to the specifications provided below.

In the specifications for both overnight and trading-hour VIX changes, lagged terms for both overnight ($\Delta \ln V_t^{ON}$) and trading-hour ($\Delta \ln V_t^{TH}$) VIX changes are omitted.

B.1.1 Baseline VAR Models (VAR1, VAR2, VAR3)

VAR1	$\Delta \ln V_t^{ON} = \alpha + \beta R_{SPF,t-1}^{TH} + e_t$ $\Delta \ln V_t^{TH} = \alpha + \beta_1 R_{SPF,t}^{ON} + \beta_2 R_{SPF,t-1}^{TH} + \beta_3 \Delta \ln V_t^{ON} + e_t$
VAR2	$\Delta \ln V_t^{ON} = \alpha + \beta R_{SPF,t-1}^{TH} + \phi_1 NTD_t + \phi_2 NTD_{t-1} + e_t$ $\Delta \ln V_t^{TH} = \alpha + \beta_1 R_{SPF,t}^{ON} + \beta_2 R_{SPF,t-1}^{TH} + \beta_3 \Delta \ln V_t^{ON} + \phi_1 NTD_t + \phi_2 NTD_{t-1} + e_t$
VAR3	$\Delta \ln V_t^{ON} = \alpha + \beta R_{SPF,t-1}^{TH} + \phi_1 NTD_t + \phi_2 NTD_{t-1} + \psi_1 D_{t+1}^{FOMC} + \psi_2 D_t^{FOMC} + \psi_3 D_{t-1}^{FOMC} + e_t$ $\Delta \ln V_t^{TH} = \alpha + \beta_1 R_{SPF,t}^{ON} + \beta_2 R_{SPF,t-1}^{TH} + \beta_3 \Delta \ln V_t^{ON} + \phi_1 NTD_t + \phi_2 NTD_{t-1} + \psi_1 D_{t+1}^{FOMC} + \psi_2 D_t^{FOMC} + \psi_3 D_{t-1}^{FOMC} + e_t$

B.1.2 Extensions with Bitcoin Overnight Returns

VAR1-ON	$\Delta \ln V_t^{ON} = \alpha + \beta R_{SPF,t-1}^{TH} + e_t$ $\Delta \ln V_t^{TH} = \alpha + \beta_1 R_{BTC,t}^{ON} + \beta_2 R_{SPF,t}^{ON} + \beta_3 R_{SPF,t-1}^{TH} + \beta_4 \Delta \ln V_t^{ON} + e_t$
VAR2-ON	$\Delta \ln V_t^{ON} = \alpha + \beta R_{SPF,t-1}^{TH} + \phi_1 NTD_t + \phi_2 NTD_{t-1} + e_t^{ON}$ $\Delta \ln V_t^{TH} = \alpha + \beta_1 R_{BTC,t}^{ON} + \beta_2 R_{SPF,t}^{ON} + \beta_3 R_{SPF,t-1}^{TH} + \beta_4 \Delta \ln V_t^{ON} + \phi_1 NTD_t + \phi_2 NTD_{t-1} + e_t$
VAR3-ON	$\Delta \ln V_t^{ON} = \alpha + \beta R_{SPF,t-1}^{TH} + \phi_1 NTD_t + \phi_2 NTD_{t-1} + \psi_1 D_{t+1}^{FOMC} + \psi_2 D_t^{FOMC} + \psi_3 D_{t-1}^{FOMC} + e_t$ $\Delta \ln V_t^{TH} = \alpha + \beta_1 R_{BTC,t}^{ON} + \beta_2 R_{SPF,t}^{ON} + \beta_3 R_{SPF,t-1}^{TH} + \beta_4 \Delta \ln V_t^{ON} + \phi_1 NTD_t + \phi_2 NTD_{t-1} + \psi_1 D_{t+1}^{FOMC} + \psi_2 D_t^{FOMC} + \psi_3 D_{t-1}^{FOMC} + e_t$

B.1.3 Extensions with Bitcoin Trading-hour Returns

VAR1-TH	$\Delta \ln V_t^{ON} = \alpha + \beta_1 R_{BTC,t-1}^{TH} + \beta_2 R_{SPF,t-1}^{TH} + e_t$ $\Delta \ln V_t^{TH} = \alpha + \beta_1 R_{BTC,t-1}^{TH} + \beta_2 R_{SPF,t}^{ON} + \beta_3 R_{SPF,t-1}^{TH} + \beta_4 \Delta \ln V_t^{ON} + e_t$
VAR2-TH	$\Delta \ln V_t^{ON} = \alpha + \beta_1 R_{BTC,t-1}^{TH} + \beta_2 R_{SPF,t-1}^{TH} + \phi_1 NTD_t + \phi_2 NTD_{t-1} + e_t$ $\Delta \ln V_t^{TH} = \alpha + \beta_1 R_{BTC,t-1}^{TH} + \beta_2 R_{SPF,t}^{ON} + \beta_3 R_{SPF,t-1}^{TH} + \beta_4 \Delta \ln V_t^{ON} + \phi_1 NTD_t + \phi_2 NTD_{t-1} + e_t$

VAR3-TH	$\Delta \ln V_t^{ON} = \alpha + \beta_1 R_{BTC,t-1}^{TH} + \beta_2 R_{SPF,t-1}^{TH} + \phi_1 NTD_t + \phi_2 NTD_{t-1} + \psi_1 D_{t+1}^{FOMC} + \psi_2 D_t^{FOMC} + \psi_3 D_{t-1}^{FOMC} + e_t$ $\Delta \ln V_t^{TH} = \alpha + \beta_1 R_{BTC,t-1}^{TH} + \beta_2 R_{SPF,t}^{ON} + \beta_3 R_{SPF,t-1}^{TH} + \beta_4 \Delta \ln V_t^{ON} + \phi_1 NTD_t + \phi_2 NTD_{t-1} + \psi_1 D_{t+1}^{FOMC} + \psi_2 D_t^{FOMC} + \psi_3 D_{t-1}^{FOMC} + e_t$
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B.1.4 Extensions with Bitcoin Close-to-Close Returns

VAR1-CC	$\Delta \ln V_t^{ON} = \alpha + \beta_1 R_{BTC,t-1}^{CC} + \beta_2 R_{SPF,t-1}^{TH} + e_t$ $\Delta \ln V_t^{TH} = \alpha + \beta_1 R_{BTC,t-1}^{CC} + \beta_2 R_{SPF,t}^{ON} + \beta_3 R_{SPF,t-1}^{TH} + \beta_4 \Delta \ln V_t^{ON} + e_t$
VAR2-CC	$\Delta \ln V_t^{ON} = \alpha + \beta_1 R_{BTC,t-1}^{CC} + \beta_2 R_{SPF,t-1}^{TH} + \phi_1 NTD_t + \phi_2 NTD_{t-1} + e_t$ $\Delta \ln V_t^{TH} = \alpha + \beta_1 R_{BTC,t-1}^{CC} + \beta_2 R_{SPF,t}^{ON} + \beta_3 R_{SPF,t-1}^{TH} + \beta_4 \Delta \ln V_t^{ON} + \phi_1 NTD_t + \phi_2 NTD_{t-1} + e_t$
VAR3-CC	$\Delta \ln V_t^{ON} = \alpha + \beta_1 R_{BTC,t-1}^{CC} + \beta_2 R_{SPF,t-1}^{TH} + \phi_1 NTD_t + \phi_2 NTD_{t-1} + \psi_1 D_{t+1}^{FOMC} + \psi_2 D_t^{FOMC} + \psi_3 D_{t-1}^{FOMC} + e_t$ $\Delta \ln V_t^{TH} = \alpha + \beta_1 R_{BTC,t-1}^{CC} + \beta_2 R_{SPF,t}^{ON} + \beta_3 R_{SPF,t-1}^{TH} + \beta_4 \Delta \ln V_t^{ON} + \phi_1 NTD_t + \phi_2 NTD_{t-1} + \psi_1 D_{t+1}^{FOMC} + \psi_2 D_t^{FOMC} + \psi_3 D_{t-1}^{FOMC} + e_t$

B.2 HAR Model Family

The HAR models directly forecast the VIX closing level. All specifications include both overnight SPX futures returns ($R_{SPF,t}^{ON}$) and lagged trading-hour SPX futures returns ($R_{SPF,t-1}^{TH}$). For notational simplicity, these terms are suppressed in the following equations.

B.2.1 Baseline HAR Model

$$\ln V_t^C = \alpha + \beta_1 \ln V_{t-1}^C + \beta_2 \ln V_{t-1}^{C(5)} + \beta_3 \ln V_{t-1}^{C(22)} + e_t$$

B.2.2 Extensions with Bitcoin Returns

HAR-ON	$\ln V_t^C = \alpha + \beta R_{BTC,t}^{ON} + \gamma_1 \ln V_{t-1}^C + \gamma_2 \ln V_{t-1}^{C(5)} + \gamma_3 \ln V_{t-1}^{C(22)} + e_t$
HAR-TH	$\ln V_t^C = \alpha + \beta R_{BTC,t-1}^{TH} + \gamma_1 \ln V_{t-1}^C + \gamma_2 \ln V_{t-1}^{C(5)} + \gamma_3 \ln V_{t-1}^{C(22)} + e_t$
HAR-CC	$\ln V_t^C = \alpha + \beta R_{BTC,t-1}^{CC} + \gamma_1 \ln V_{t-1}^C + \gamma_2 \ln V_{t-1}^{C(5)} + \gamma_3 \ln V_{t-1}^{C(22)} + e_t$

Appendix C. Out-of-Sample Test with Alternative Estimation Window

This table evaluates the robustness of out-of-sample forecasting performance using an alternative estimation window. Models exhibiting superior performance are highlighted in grey. Panel A assesses forecasting performance using four evaluation metrics: Mean Squared Error (MSE), Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), and the coefficient of determination (R^2). For comparability, MSE and MAPE values are multiplied by 100, and MAE values are multiplied by 10. Panel B presents Diebold-Mariano (DM) test statistics, comparing the forecasting accuracy of competing models using MAE as the loss function. A significantly positive (negative) DM test statistic indicates that the row model has superior (inferior) forecasting performance relative to the column model. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. Panel C reports results from the Model Confidence Set (MCS) test. For each model, p -values are provided for four loss metrics: MSE, heteroskedasticity-adjusted MSE (HMSE), MAE, and MAPE. Models with p -values marked by * are included in the 90% confidence set, and those marked by ** are included in the 75% confidence set. The sample period spans January 2018 to December 2023, with model parameters estimated using a rolling window of 750 observations. Models exhibiting superior performance are highlighted in grey.

Panel A: Average Forecasting Losses				
	MSE	MAPE	MAE	R^2
HAR	0.518	1.739	0.527	0.951
HAR-ON	0.500	1.714	0.519	0.953
HAR-TH	0.519	1.741	0.528	0.951
HAR-CC	0.519	1.742	0.528	0.951
VAR1	0.549	1.759	0.535	0.949
VAR1-ON	0.534	1.740	0.529	0.950
VAR1-TH	0.550	1.760	0.535	0.948
VAR1-CC	0.550	1.760	0.535	0.948
VAR2	0.539	1.729	0.526	0.949
VAR2-ON	0.524	1.707	0.519	0.951
VAR2-TH	0.540	1.730	0.527	0.949
VAR2-CC	0.540	1.730	0.527	0.949
VAR3	0.537	1.725	0.525	0.950
VAR3-ON	0.521	1.702	0.517	0.951
VAR3-TH	0.538	1.724	0.525	0.950
VAR3-CC	0.538	1.726	0.526	0.950
Panel B: Diebold-Mariano Test Results				
	HAR-ON	VAR1-ON	VAR2-ON	VAR3-ON
HAR	-2.32**			
HAR-TH	-2.53**			
HAR-CC	-2.45**			
VAR1		-2.36**		
VAR1-TH		-2.41**		
VAR1-CC		-2.50**		
VAR2			-2.51**	
VAR2-TH			-2.58***	
VAR2-CC			-2.57**	
VAR3				-2.60***
VAR3-TH				-2.56**
VAR3-CC				-2.67***

Panel C: Model Confidence Set (MCS)				
	<i>MSE</i>	<i>HMSE</i>	<i>MAE</i>	<i>MAPE</i>
<i>HAR</i>	0.080	0.059	0.192 [*]	0.242 [*]
<i>HAR-ON</i>	1.000 ^{**}	1.000 ^{**}	1.000 ^{**}	1.000 ^{**}
<i>HAR-TH</i>	0.062	0.042	0.129 [*]	0.168 [*]
<i>HAR-CC</i>	0.066	0.050	0.142 [*]	0.186 [*]
<i>VAR1</i>	0.020	0.027	0.098	0.090
<i>VAR1-ON</i>	0.080	0.108 [*]	0.380 ^{**}	0.334 ^{**}
<i>VAR1-TH</i>	0.024	0.031	0.071	0.068
<i>VAR1-CC</i>	0.018	0.024	0.092	0.087
<i>VAR2</i>	0.033	0.030	0.119 [*]	0.171 [*]
<i>VAR2-ON</i>	0.587 ^{**}	0.785 ^{**}	0.977 ^{**}	0.963 ^{**}
<i>VAR2-TH</i>	0.044	0.045	0.125 [*]	0.177 [*]
<i>VAR2-CC</i>	0.018	0.017	0.100	0.151 [*]
<i>VAR3</i>	0.027	0.025	0.097	0.143 [*]
<i>VAR3-ON</i>	0.683 ^{**}	0.875 ^{**}	1.000 ^{**}	1.000 ^{**}
<i>VAR3-TH</i>	0.044	0.049	0.149 [*]	0.211 [*]
<i>VAR3-CC</i>	0.015	0.014	0.080	0.129 [*]

Appendix D. Performance of VXX-Based Trading Strategies

This table evaluates the economic performance of long-short trading strategies using VXX over the sample period from January 2018 to December 31, 2023. Model parameters are estimated using a rolling window covering one-third of the sample. Performance is assessed using multiple metrics: average daily abnormal returns (in percentage), the standard deviation of daily abnormal returns, the annualized Information Ratio (IR), and annualized certainty equivalent return (CER) gains relative to a passive buy-and-hold strategy. The CER is computed as $CER = \mu - 0.5\gamma\sigma^2$, where μ and σ^2 represent the mean and variance of annualized asset returns, and γ denotes the relative risk aversion coefficient of a mean-variance investor. We consider $\gamma = 1$ for low-risk version and $\gamma = 3$ for high-risk aversion. CER gains measure the difference between the CER of the long-short strategy and that of the passive buy-and-hold strategy. Models showing superior performance are highlighted in grey.

	Average daily abnormal return (%)	Std of daily abnormal return	Annualized I.R.	Annualized CER gains ($\gamma=1$)	Annualized CER gains ($\gamma=3$)
<i>HAR</i>	0.241	0.097	0.405	0.255	-0.521
<i>HAR-ON</i>	0.523	0.095	0.894	1.048	0.362
<i>HAR-TH</i>	0.264	0.097	0.442	0.314	-0.460
<i>HAR-CC</i>	0.210	0.097	0.353	0.165	-0.620
<i>VAR1</i>	0.020	0.095	0.035	-0.282	-0.955
<i>VAR1-ON</i>	0.011	0.095	0.020	-0.295	-0.947
<i>VAR1-TH</i>	-0.018	0.095	-0.032	-0.388	-1.067
<i>VAR1-CC</i>	-0.026	0.095	-0.045	-0.394	-1.037
<i>VAR2</i>	0.145	0.091	0.261	0.162	-0.290
<i>VAR2-ON</i>	0.178	0.090	0.322	0.258	-0.170
<i>VAR2-TH</i>	0.153	0.091	0.274	0.168	-0.311
<i>VAR2-CC</i>	0.151	0.091	0.271	0.173	-0.285
<i>VAR3</i>	0.103	0.091	0.184	0.043	-0.418
<i>VAR3-ON</i>	0.155	0.088	0.288	0.250	-0.080
<i>VAR3-TH</i>	0.147	0.093	0.259	0.112	-0.447
<i>VAR3-CC</i>	0.130	0.092	0.230	0.080	-0.455